

Multi-Faceted Customer Intelligence

An End-to-End Data Mining Pipeline on Online Retail II

Data Mining 2026 — Final Project

Team 4 — University of Information Technology, VNU-HCM

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The Problem

Three siloed questions every retailer asks

- WHO are my customers? — segmentation
- WHO will leave? — churn prediction
- WHAT do they buy together? — basket analysis

Solved in isolation, these analyses miss the joint signal that drives retention ROI.

Why this matters

+5%

customer retention

drives 25–95% profit growth
(Bain & Co.)

Dataset — UCI Online Retail II

UK-centric online retailer, 2009-12 to 2011-12

1.07M

raw transactions

805K

after cleaning

5,878

unique customers

4,631

stock codes

41

countries

2 yr

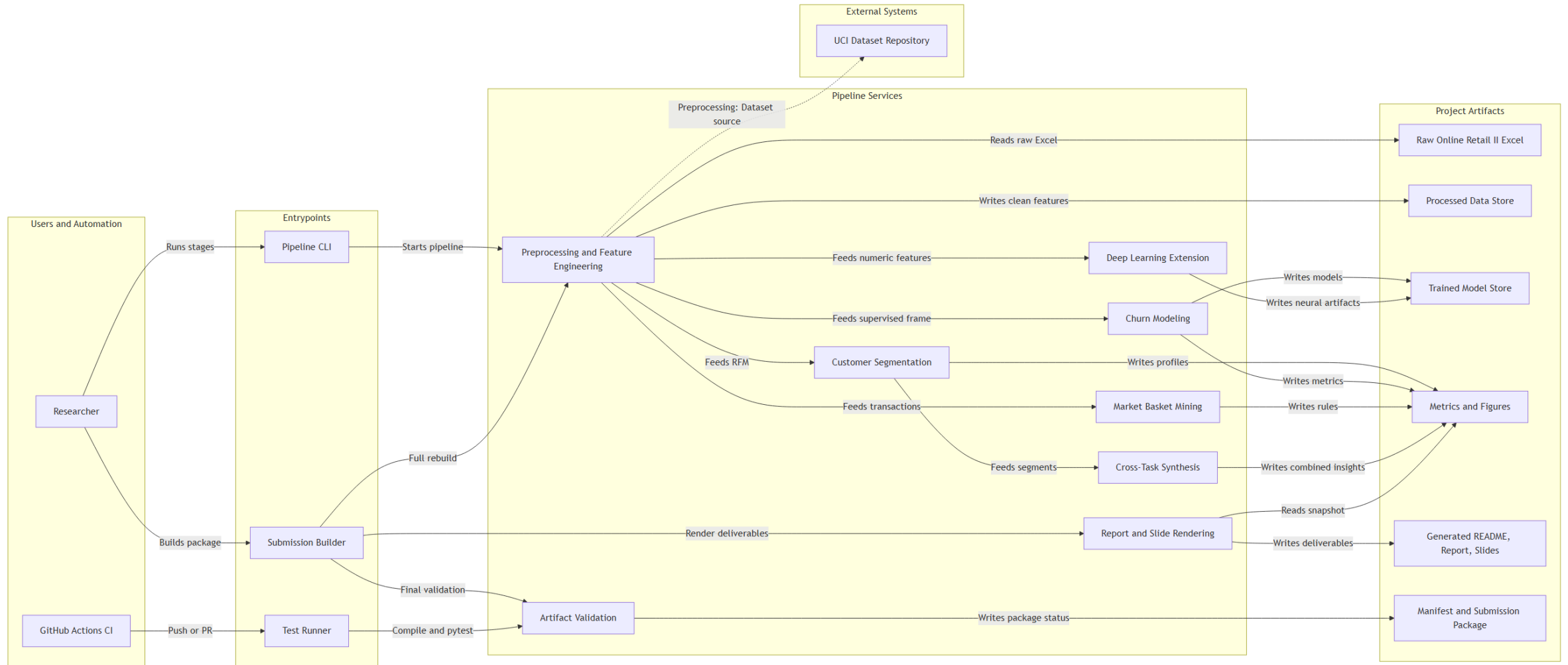
time span

Columns: Invoice · StockCode · Description · Quantity · Price · InvoiceDate · CustomerID · Country

Source: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>

System Architecture

Users & automation → entry points → pipeline services → artifact stores

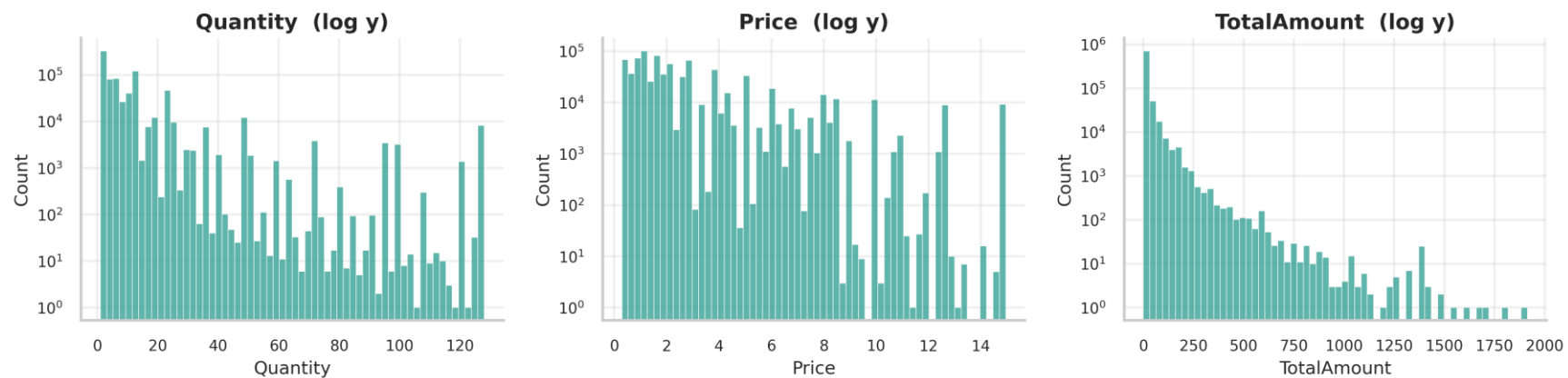


Preprocessing

From raw 1.07M rows to clean 805K rows

- Drop ~25% rows with null CustomerID
- Remove cancellation invoices (prefix 'C')
- Drop non-positive Quantity / Price
- Winsorize Quantity & Price at 1%/99% quantiles
- Derive TotalAmount = Quantity × Price
- Compute RFM + behavioral features per customer
- Label churn: no activity in 90 d after 2011-09-01

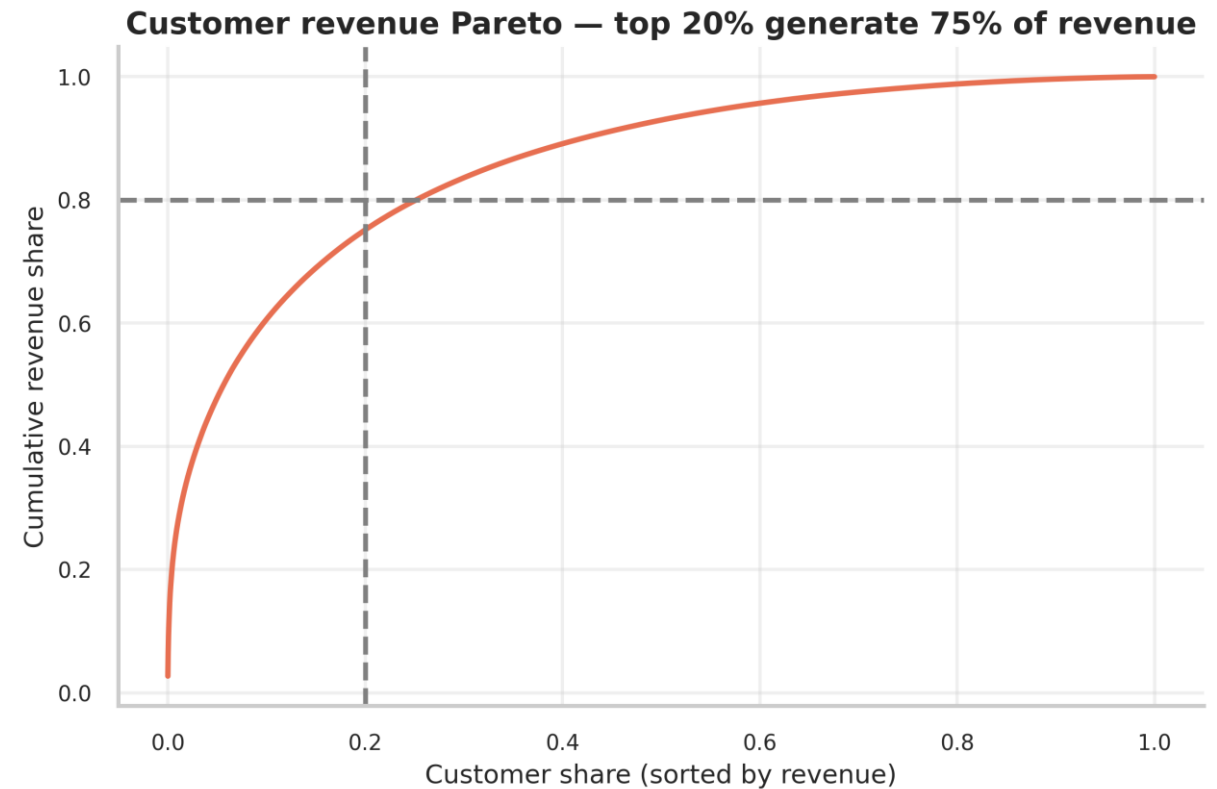
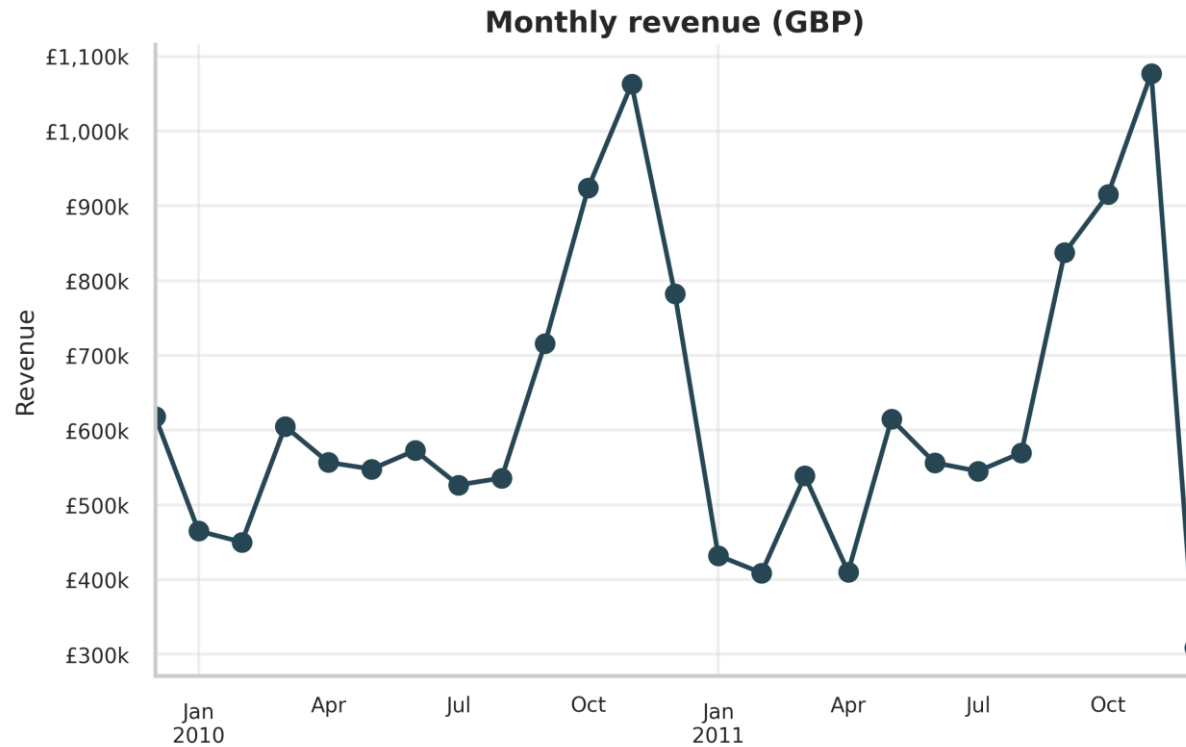
Distributions after winsorization



Quantity / Price / TotalAmount distributions after winsorization

EDA — Revenue & Customer Pareto

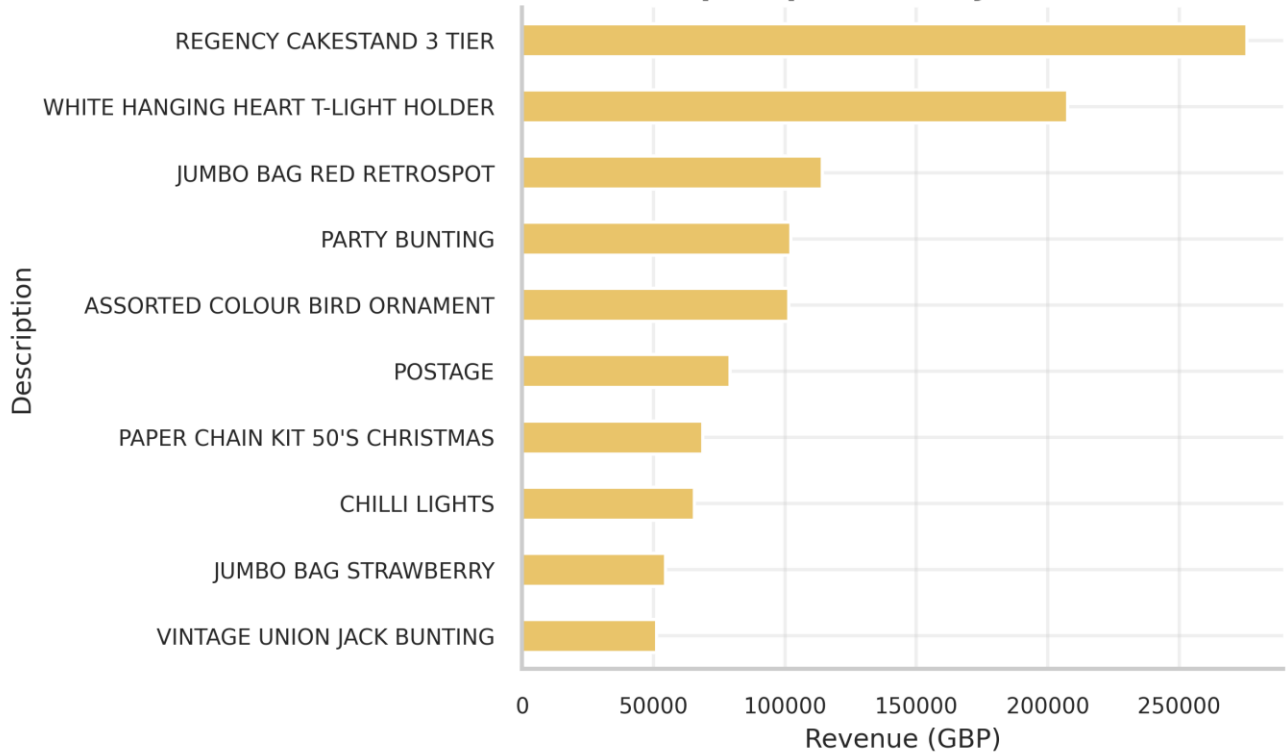
Q4 seasonality + top 20% of customers drive most revenue



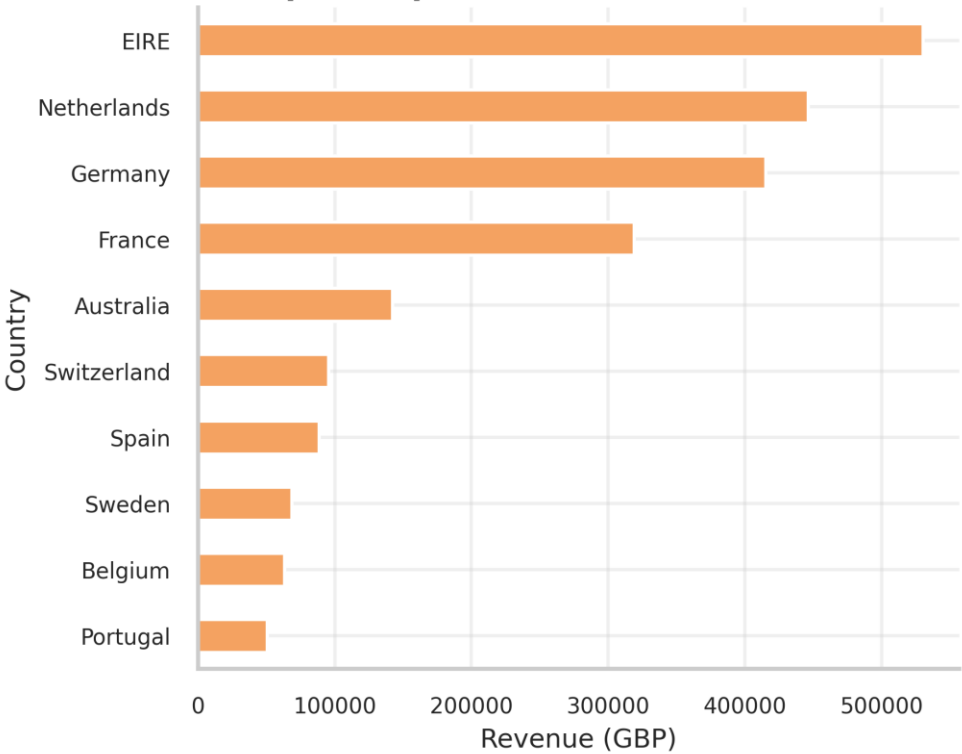
EDA — Top Products & Export Countries

Long-tail product mix; concentrated international demand

Top 10 products by revenue



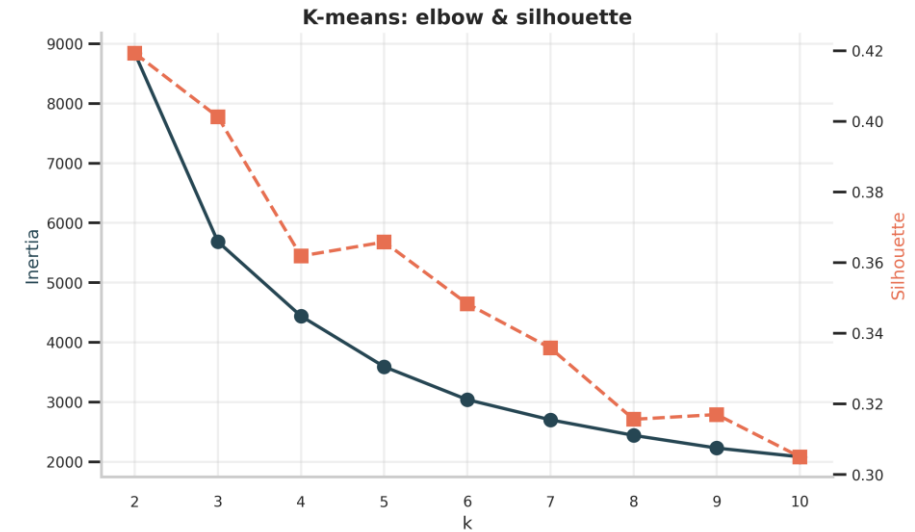
Top 10 export countries (UK excluded)



Customer Segmentation — Three Algorithms Compared

K-means vs DBSCAN vs AGNES on log-transformed RFM

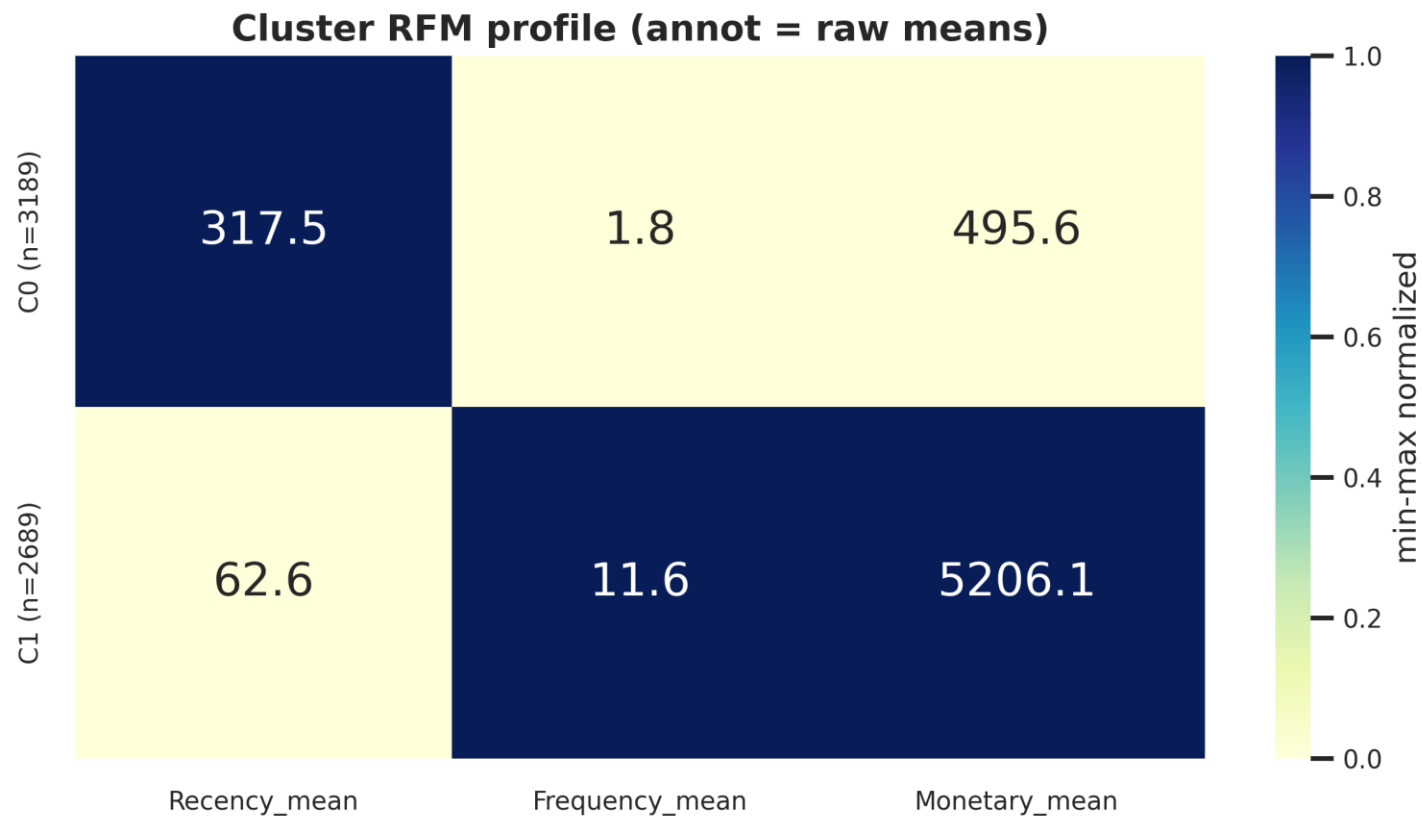
Algorithm	k	Silhouette ↑	Davies-B ↓	Calinski-H ↑
KMeans	2	0.419	0.886	5840
DBSCAN	13	-0.120	1.632	742
AGNES	2	0.388	0.789	4304



- K-means k=2 wins silhouette (0.42) — best partitional fit
- DBSCAN fragments into 13 + noise — wrong fit for heavy-tail
- AGNES Ward close second; chosen k=2 confirmed by dendrogram

Two Customer Segments — Crystal Clear Interpretation

Dormant (54%) vs Active loyal (46%)



Dormant

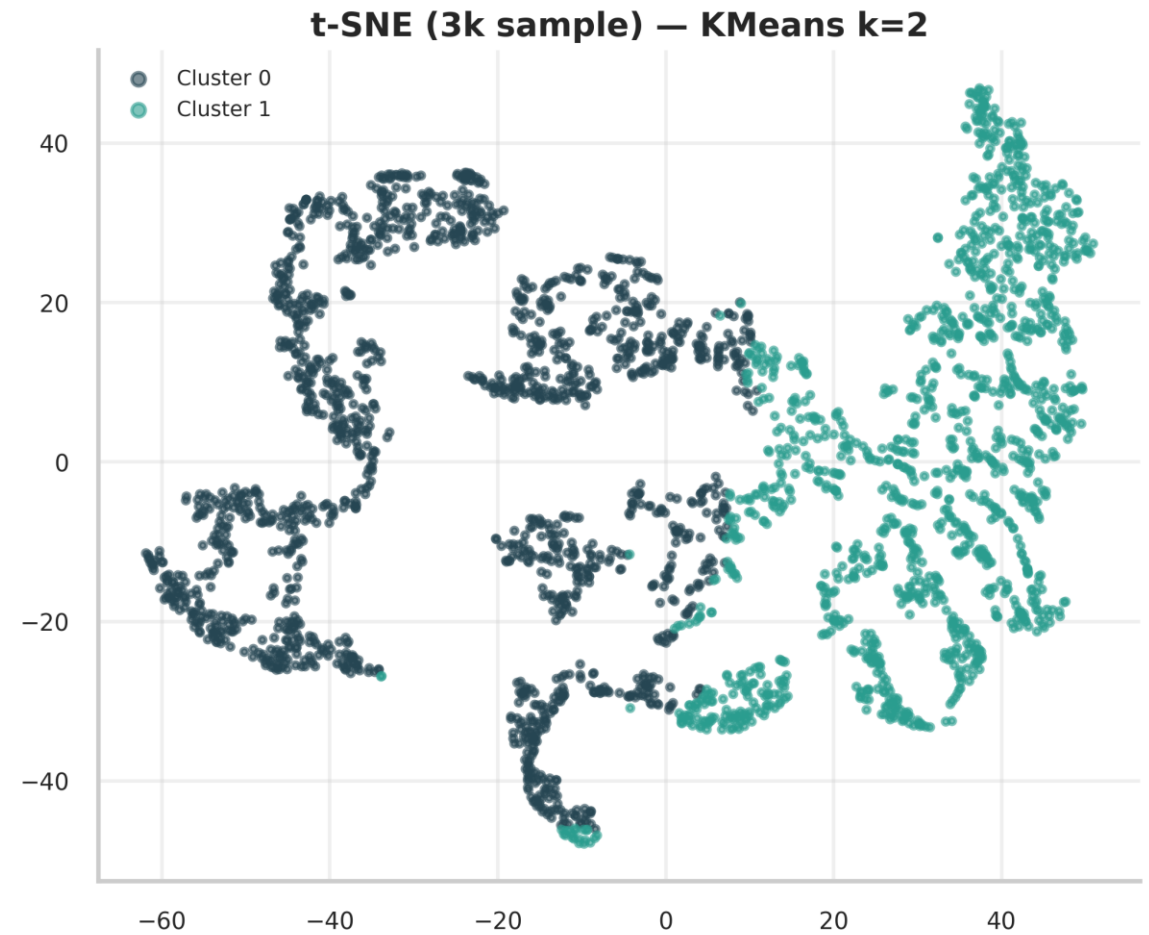
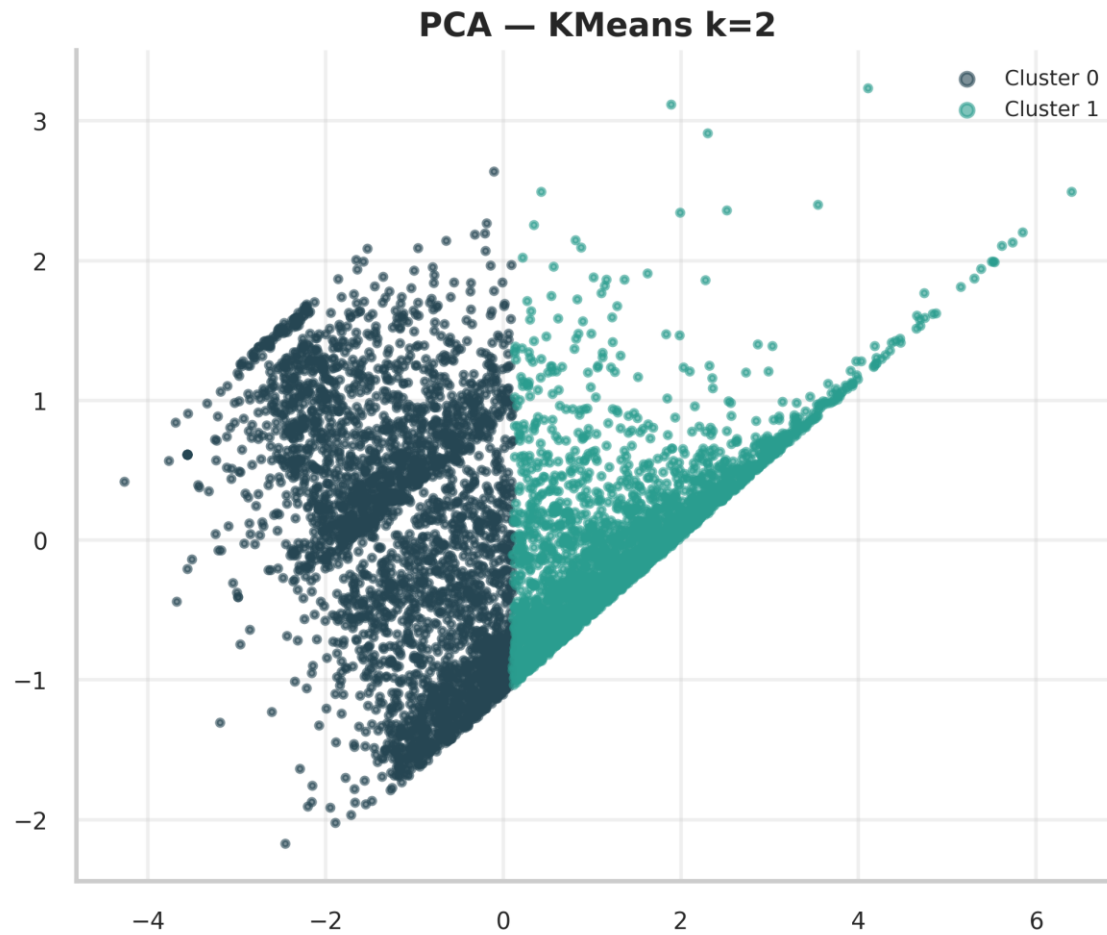
Recency 317 d
Frequency 1.85
Monetary £496

Active loyal

Recency 63 d
Frequency 11.55
Monetary £5,206

2-D Projection — PCA & t-SNE

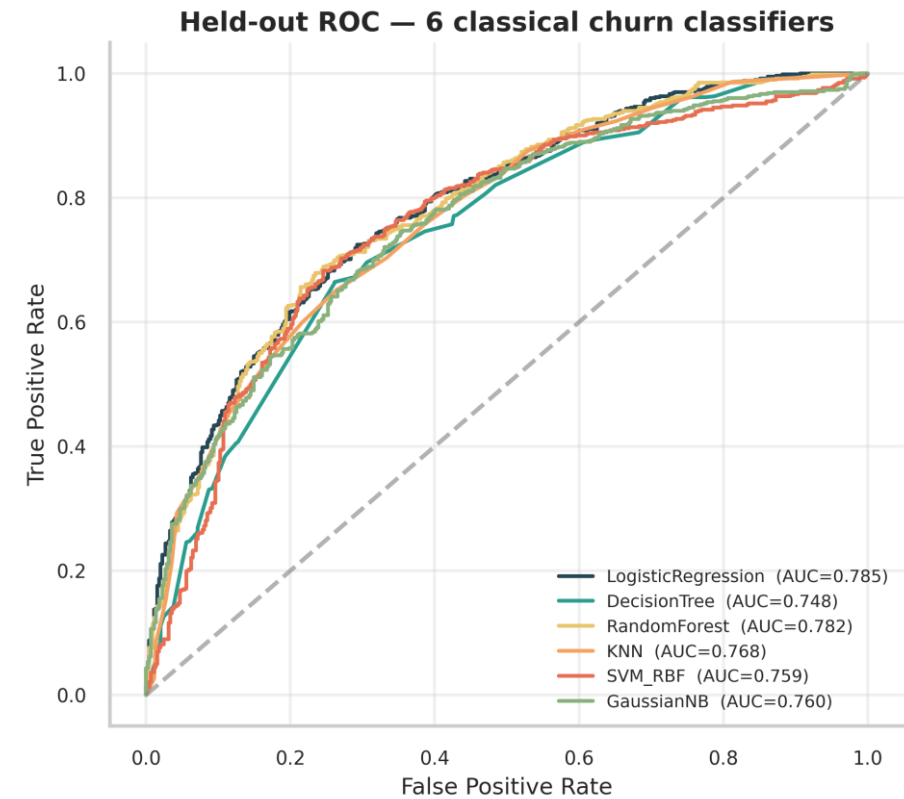
Clean separation along the first principal component



Churn Classification — 7 Models Compared

5-fold CV + held-out test; SMOTE inside folds

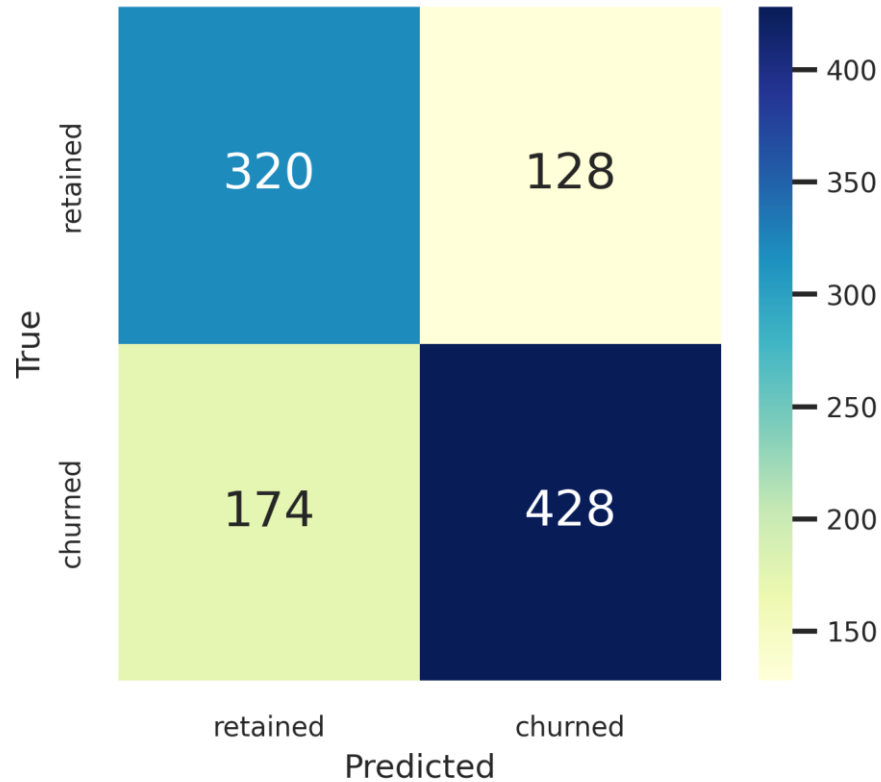
Model	AUC	F1	Precision	Recall
LogisticRegression	0.785	0.739	0.770	0.711
RandomForest	0.782	0.745	0.747	0.744
KNN	0.767	0.721	0.740	0.703
GaussianNB	0.760	0.744	0.724	0.766
SVM_RBF	0.759	0.745	0.752	0.738
DecisionTree	0.748	0.724	0.754	0.696
MLP (PyTorch)	0.783	0.738	0.756	0.721



Best Model — LogisticRegression (AUC 0.785)

RandomForest within 0.003 · MLP AUC 0.783

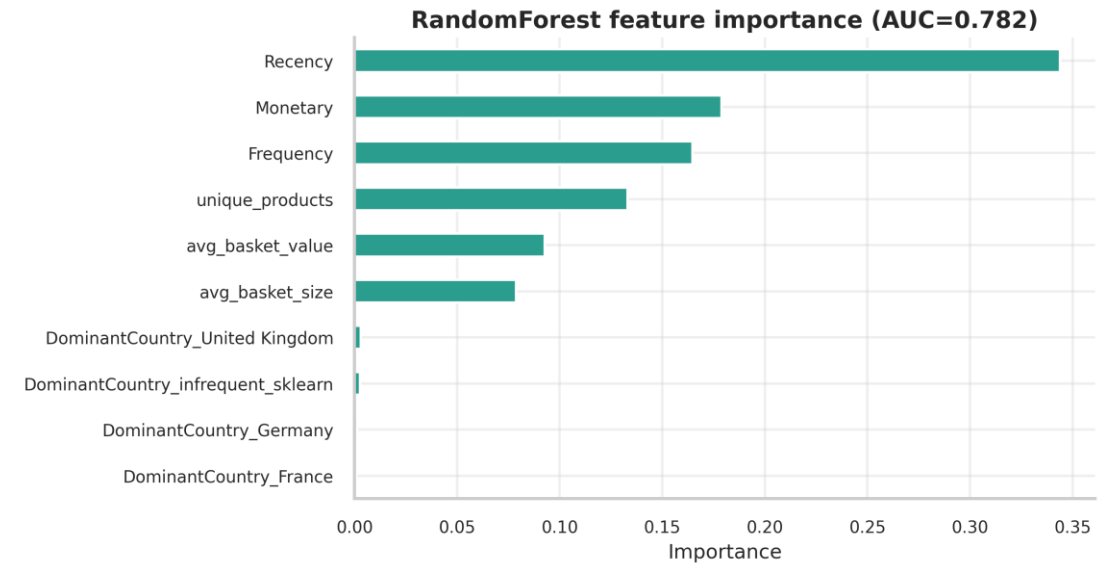
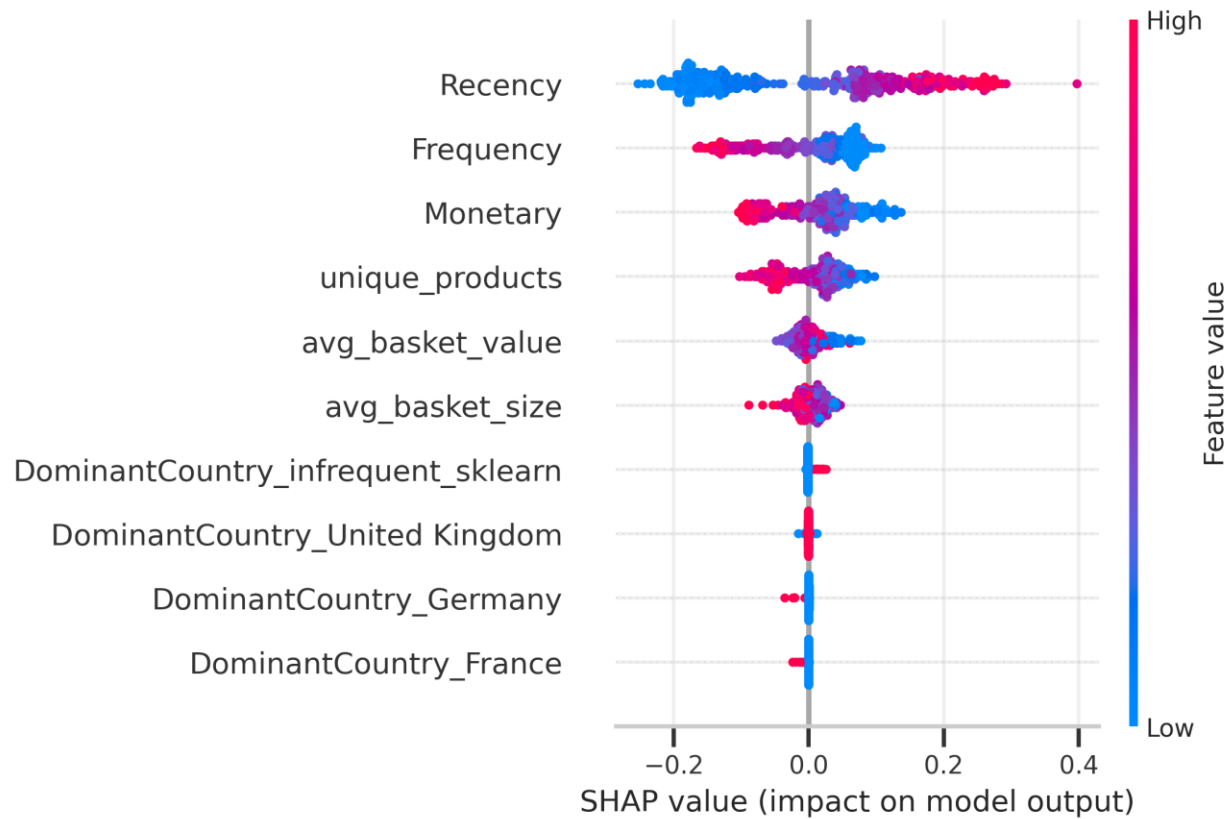
Best classical: LogisticRegression (AUC=0.785)



- AUC 0.7850 · F1 0.7392 · Recall 0.7110
- MLP test: AUC 0.7832 · Precision 0.7561 · Recall 0.7209
- Classical and MLP converge — feature ceiling on RFM+behavioural
- → classical models preferred (cheaper, interpretable)
- → DL would need richer signals (sequences, sessions) to pull ahead

Explainability — SHAP on RandomForest

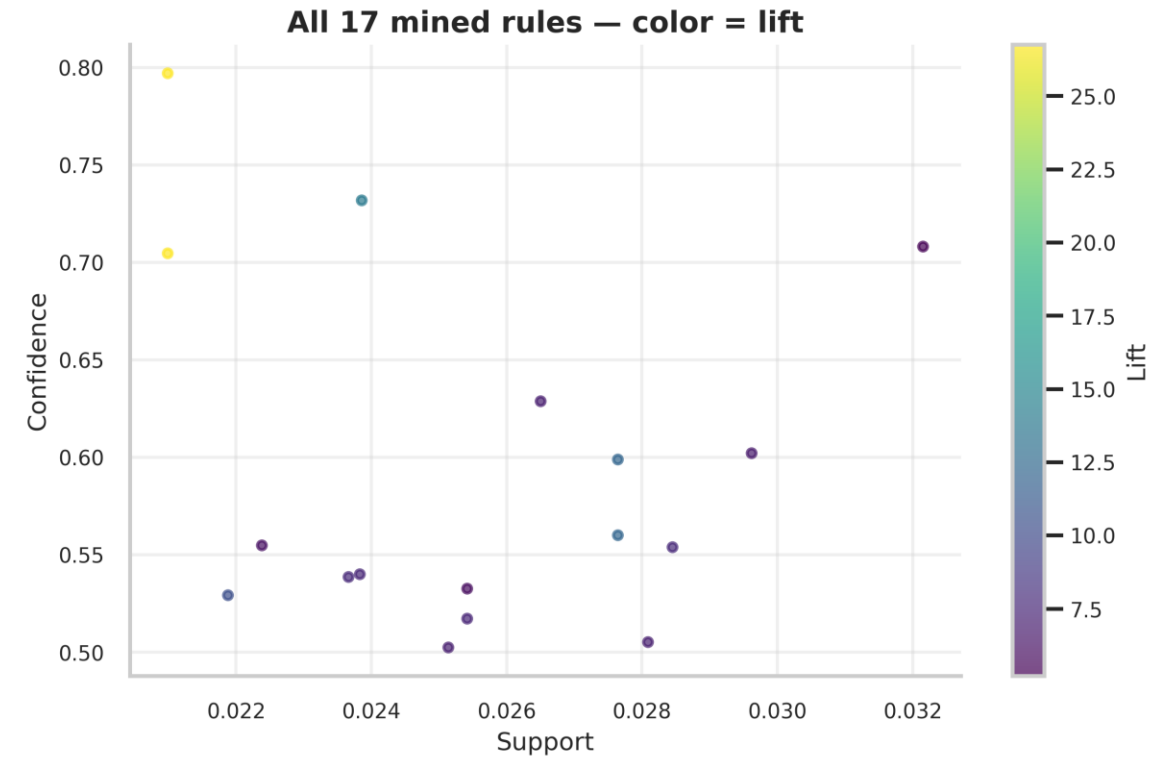
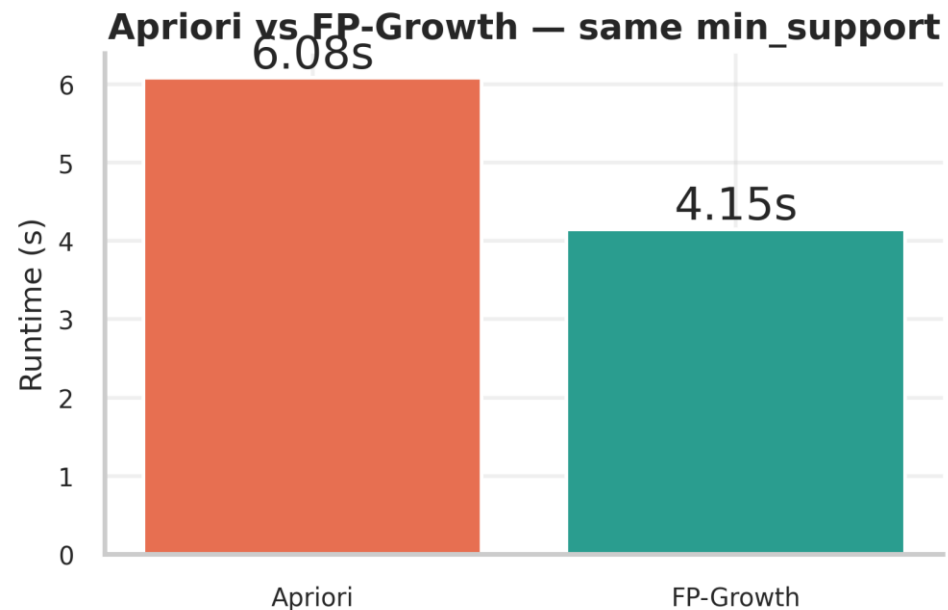
Recency dominates; behavioural features add the rest



Market Basket — Apriori vs FP-Growth

Same 242 itemsets, 17 rules at lift ≥ 1.5 — FP-Growth 1.5× faster

Algorithm	Itemsets	Rules	Runtime (s)
Apriori	242	17	6.56
FP-Growth	242	17	4.15



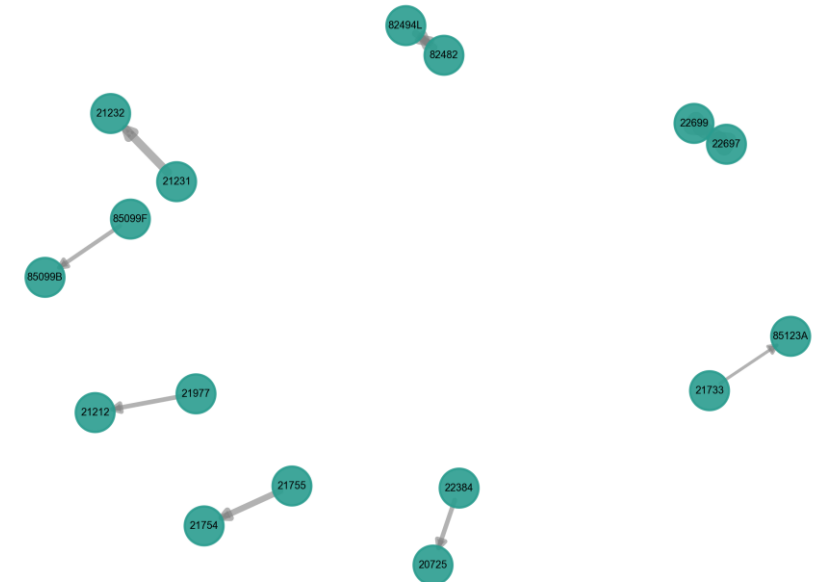
Top Association Rules

Strongest signals = collectible sets + color-variant pairs

Antecedent	Consequent	Conf.	Lift
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.70	26.8
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.80	26.8
SWEETHEART CERAMIC TRINKET BOX	STRAWBERRY CERAMIC TRINKET BOX	0.73	13.9
WOODEN FRAME ANTIQUE WHITE	WOODEN PICTURE FRAME WHITE FINISH	0.56	12.1
WOODEN PICTURE FRAME WHITE FINISH	WOODEN FRAME ANTIQUE WHITE	0.60	12.1

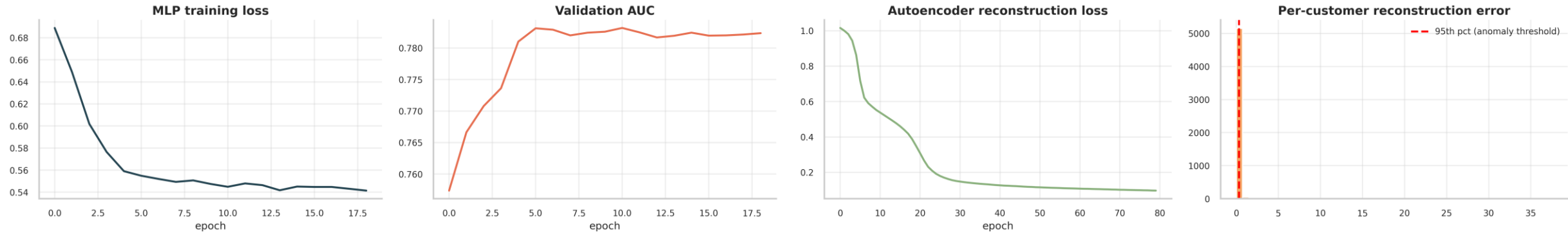
Pattern: customers who buy one item from a colour or themed set frequently complete the set. Operational use: post-purchase recommendations and bundled retention offers.

Top-20 association rules (edge width = lift)



Deep Learning Extension

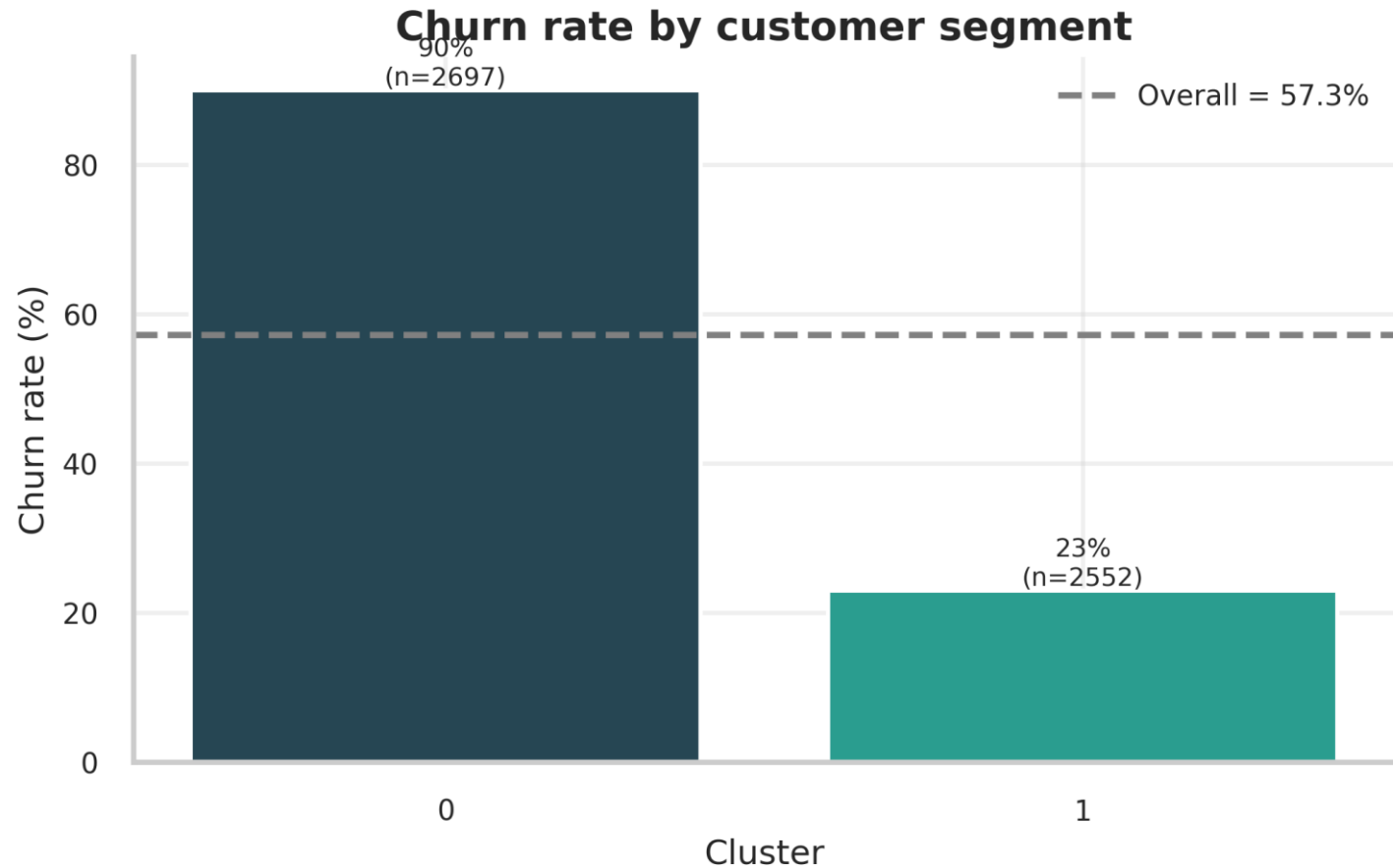
MLP for churn + Autoencoder for anomaly detection



MLP [64,32,16] with SMOTE + early stopping reaches test AUC 0.783 (F1 0.738, precision 0.756) — on par with the best classical model. Autoencoder (bottleneck=4) flags the top 5% of customers by reconstruction error as anomalies — candidates for VIP review or fraud screening.

★ HEADLINE FINDING — Churn Rate by Segment

67-percentage-point gap between segments



Dormant

90%

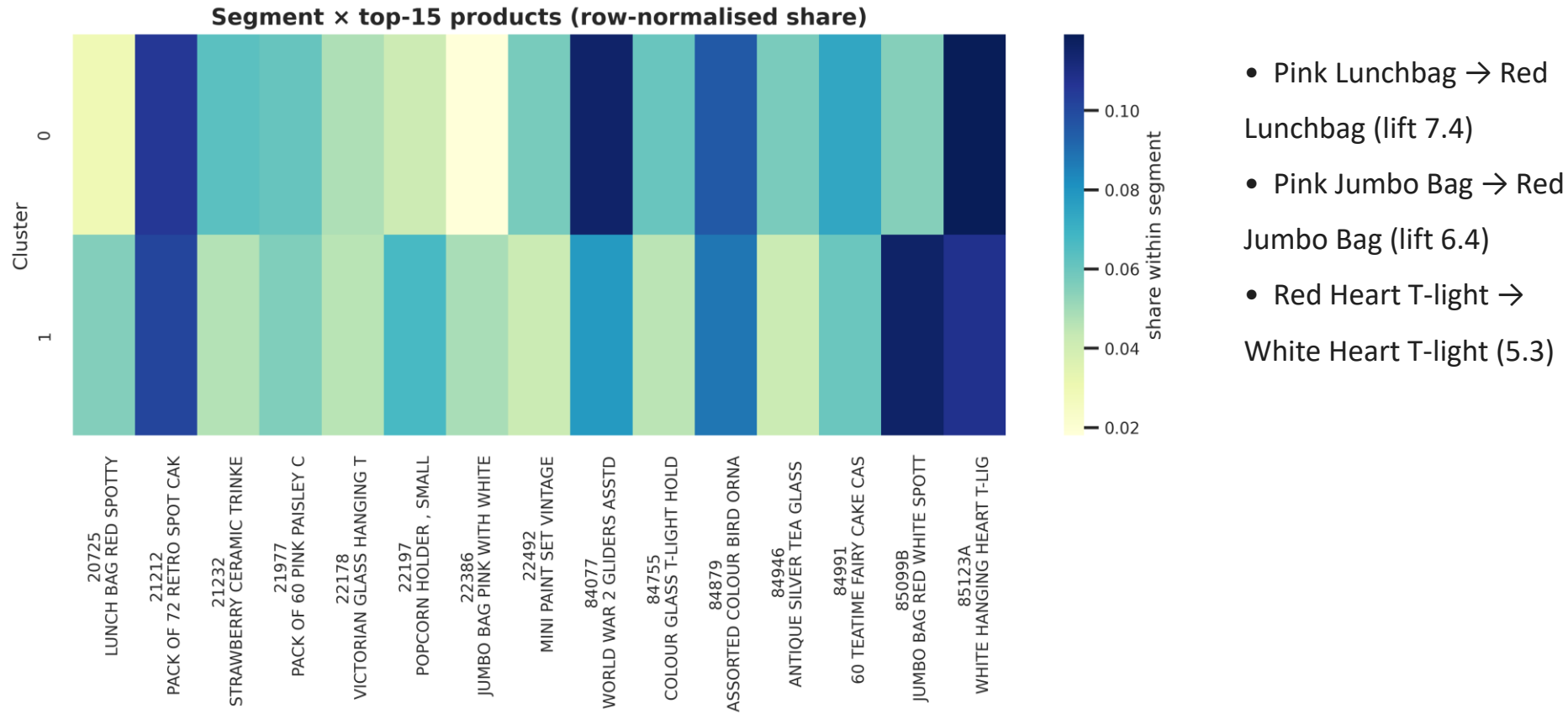
Active loyal

23%

67-point churn gap
→ segment-aware retention works

Loyal Segment Buys in Colour Pairs

Per-segment FP-Growth recovers cross-sell opportunities



Recommendation: target the 23% churners in the loyal segment with colour-pair bundles. Use win-back campaigns (not bundles) for the dormant majority.

Limitations & Future Work

Honest caveats — and where we'd go next

Limitations

- Leakage fix: supervised features built strictly before 2011-09-01 — best AUC now a realistic 0.785 (full-history features were inflated)
- Recency dominant but not sufficient — ablation without it drops AUC to 0.75; baselines: majority 0.50, recency-only 0.75
- Single retailer, UK-centric, 2 yrs — generalisability untested
- No demographic attributes available
- Dormant segment too sparse for in-cluster rules
- Anomaly threshold purely percentile-based

Future Work

- Transformer on purchase sequences (next-basket + churn jointly)
- Uplift modelling: who responds best to retention offers
- Cross-retailer transfer to test segment-rule stability
- Domain-labelled fraud / VIP ground truth for anomalies

Conclusion & Thank You

Q&A — happy to discuss any slide

One pipeline, four course topics, one business insight.

- Segmentation: K-means k=2, Silhouette 0.42
- Classification: LogisticRegression AUC 0.785 · MLP AUC 0.783 (leakage-safe pre-cutoff features)
- Association: 17 rules, FP-Growth 1.5× faster, top lift 26.8
- ★ Synthesis: 67-pt churn gap → colour-pair retention bundles

Code, report & slides — Team 4 GitHub repository

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